Question 1:

Find a solution of the differential equation $\left(3y+\sqrt{x^2-y^2}\right)dx = 3x\cdot dy$ or $\dfrac{dy}{dx}=\dfrac{3y+\sqrt{x^2-y^2}}{3x}$

a) Show that the function $f(x)=\dfrac{3y+\sqrt{x^2-y^2}}{3x}$ is homogenous.

b) The general solution of the differential equation $\left(3y+\sqrt{x^2-y^2}\right)dx = 3x\cdot dy$ is ______?______

Hint: Formula 18

$$\int \frac{1}{\sqrt{a^2-u^2}}\,du = \arcsin\left(\frac{u}{a}\right) + C$$

Question 2:

Find a solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}$ or $x \cdot dy = (x+y)dx$

a) Show that the function $f(x) = \frac{x+y}{x}$ is homogenous.

b) The general solution of the differential equation $x \cdot dy = (x+y)dx$ is _______?_______

Question 3:

Find a solution of the differential equation $\frac{dy}{dx} = \frac{y^2}{xy + 2x^2}$ or $(xy + 2x^2) \cdot dy = y^2 dx$

a) Show that the function $f(x) = \frac{y^2}{xy + 2x^2}$ is homogenous.

b) The general solution of the differential equation $(xy + 2x^2) \cdot dy = y^2 dx$ is ______?______

Question 4:

Find a solution of the differential equation $\frac{dy}{dx}=\frac{2x^3-4x^2y}{x^3}$ or $x^3\cdot dy=\left(2x^3-4x^2y\right)dx$

a) Show that the function $f(x)=\frac{2x^3-4x^2y}{x^3}$ is homogenous.

b) The general solution of the differential equation $x^3\cdot dy=\left(2x^3-4x^2y\right)dx$ is _____?_____

Question 5:

Find a solution of the differential equation $\frac{dy}{dx} = \frac{x^2 - y^2}{4xy}$ or $4xy \cdot dy = \left(x^2 - y^2\right)dx$

a) Show that the function $f(x) = \frac{x^2 - y^2}{4xy}$ is homogenous.

b) The general solution of the differential equation $4xy \cdot dy = \left(x^2 - y^2\right)dx$ is ______?______